

FORT BEAUFORT, South Africa

WMO: 687470

Lat: 32.788S

Lon: 26.629E

Elev: 1493

StdP: 13.92

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	36.8	39.1	17.7	14.2	73.9	22.4	17.8	67.0	24.6	62.6	22.3	72.2	4.1	300	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	96.0	67.6	92.0	67.3	88.6	66.7	72.6	86.3	71.1	84.1	70.0	82.5	9.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.5	110.9	77.8	67.3	106.2	76.3	66.2	101.9	75.1	37.2	86.6	35.9	84.2	34.9	82.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.4	16.9	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	74.0	73.8	70.9	65.7	61.7	57.1	57.7	59.0	61.5	64.9	67.7	71.5
	DBStd	8.61	6.09	5.42	5.91	6.41	6.01	6.47	7.17	6.78	6.26	6.57	6.25	6.02
	HDD50	23	0	0	0	0	1	6	10	4	2	0	0	0
	HDD65	1227	6	4	13	66	139	252	249	216	142	84	43	13
	CDD50	5651	744	666	648	472	365	219	249	282	345	463	532	666
	CDD65	1382	285	250	196	88	38	15	24	29	36	82	125	214
	CDH74	10781	2262	1733	1326	645	357	150	225	367	373	732	964	1647
	CDH80	4135	1006	706	517	215	87	21	30	113	130	262	353	695
Wind	WSAvg	6.0	6.6	6.0	5.4	4.8	5.1	6.4	6.3	6.1	6.0	6.4	6.7	6.7
Precipitation	PrecAvg	20.4	2.0	2.6	2.8	1.5	0.8	0.7	0.6	1.2	1.2	2.0	2.5	2.5
	PrecMax	33.3	5.2	8.9	11.3	4.8	3.2	3.3	3.5	8.5	4.1	5.2	8.0	8.0
	PrecMin	9.8	0.1	0.4	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0
	PrecStd	5.9	1.4	1.9	2.4	1.1	0.8	0.8	0.9	1.5	1.1	1.3	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.1	98.2	97.9	91.7	87.1	82.4	83.6	89.4	92.2	95.0	96.0	99.2
		MCWB	69.6	69.1	67.1	64.6	60.0	56.5	54.6	56.3	58.6	63.9	65.3	68.0
	2%	DB	94.5	92.2	91.2	86.4	82.2	78.0	79.6	81.9	84.1	87.8	88.8	91.8
		MCWB	70.0	70.2	67.4	63.9	58.5	54.8	54.2	54.9	59.1	63.5	64.1	67.2
	5%	DB	89.5	88.0	86.0	81.4	77.9	74.5	76.0	77.3	77.9	82.5	83.6	87.2
		MCWB	68.5	70.2	67.0	63.4	57.6	53.2	52.8	54.2	58.9	62.8	64.2	67.4
	10%	DB	85.2	83.9	81.7	76.8	73.6	70.2	71.6	72.0	73.2	77.6	79.4	82.7
		MCWB	68.1	69.0	66.4	61.8	56.8	52.5	51.9	53.5	57.6	61.4	63.3	66.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.8	75.2	72.1	68.9	64.2	59.4	57.6	61.7	65.4	68.5	70.1	72.6
		MCDB	90.3	88.2	85.6	80.1	75.6	71.4	72.7	70.1	78.3	85.2	86.1	88.8
	2%	WB	72.1	73.0	70.3	66.9	61.8	56.6	56.2	59.2	62.9	66.6	67.7	70.6
		MCDB	86.5	85.3	83.5	77.9	74.0	70.8	71.4	70.1	75.7	80.7	82.4	85.8
	5%	WB	70.8	71.4	68.9	65.4	60.2	55.2	54.9	57.5	61.2	64.5	66.0	68.9
		MCDB	84.1	82.8	81.3	76.6	71.6	68.6	69.5	68.5	73.2	76.6	78.3	82.9
	10%	WB	69.5	70.1	67.5	63.8	59.0	53.8	53.6	55.9	59.5	62.8	64.3	67.5
		MCDB	82.3	80.7	78.9	74.0	70.4	66.7	66.9	67.3	70.7	74.0	75.1	79.7
Mean Daily Temperature Range	5% DB	MDBR	21.2	19.9	19.7	20.8	22.0	22.6	22.5	22.2	21.5	21.3	21.4	20.5
		MCDBR	29.3	27.1	27.7	27.0	28.4	28.0	27.1	31.3	30.8	30.5	30.0	29.3
		MCWBR	9.2	8.9	9.1	9.6	10.5	11.2	10.2	11.8	12.0	11.6	11.8	10.4
	5% WB	MCDBR	25.4	24.2	23.4	23.3	23.2	23.4	23.3	23.0	25.8	25.5	26.5	26.2
		MCWBR	9.1	9.3	9.6	9.9	10.8	11.2	11.3	10.5	11.5	11.0	11.5	10.4
Clear-Sky Solar Irradiance	taub		0.369	0.371	0.356	0.328	0.310	0.295	0.298	0.344	0.372	0.383	0.363	0.366
	taud		2.465	2.465	2.482	2.531	2.550	2.560	2.542	2.405	2.325	2.340	2.437	2.467
	Ebn at Noon		307	300	293	285	275	272	276	275	283	293	307	309
	Edh at Noon		37	36	33	28	25	23	24	31	38	41	38	38
All-Sky Solar Radiation	RadAvg		2172	1982	1680	1319	1046	929	1001	1244	1600	1878	2132	2265
	RadStd		113	101	119	70	42	52	56	80	109	133	123	153

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.63	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-153	+256	+81

Nomenclature: See separate page